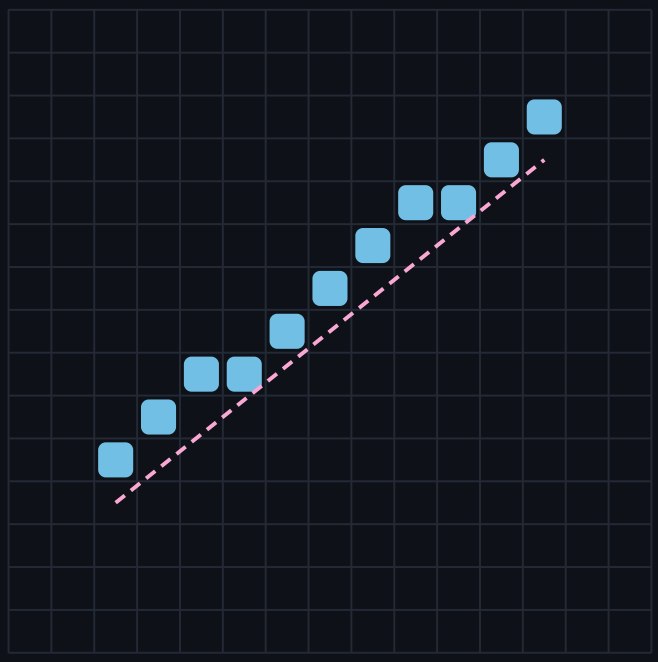
CMPN322 · DR. MANAR DONIA · FORMULA SHEET

Computer Graphics Formula Sheet

Every law you need to solve the problems — colors and memory, window to viewport, aspect ratio, DDA, Bresenham, midpoint circle and ellipse, and the 2D / 3D transformation matrices — each with a solved example from the references.

13 law cards | Solved example under every law | Lec 2–6 · Sec 3–7



Bresenham (20,10) → (30,18)



01 Color, LUT & Frame Buffer

02 Window → Viewport & Aspect Ratio

03 DDA & Bresenham

04 Midpoint Circle & Ellipse

Contents

Laws for the problems only

01 **Color, LUT & Frame Buffer** — Lec 2 · SEC 3
gray levels · direct coding · bit planes · LUT entries and size · frame buffer memory

02 **Window → Viewport & Aspect Ratio** — Lec 3 · Sec 4
Sx, Sy · world ↔ viewport · matching viewport

03 **Line Drawing — DDA & Bresenham** — Lec 4 · sec5
slope · DDA in all directions · rounding · decision parameter

04 **Midpoint Circle & Ellipse** — Lec 5 · Sec 6
P0 · updates · stop · symmetry · region 1 / region 2

05 **2D / 3D Transformations** — Lec 6 · Sec 7
equations · matrices · inverses · concatenation · composite

How to use it

- Each card: the law, the formula, and when to use it.
- The grey box under it is a solved example from the references.
- Red boxes mark numbers printed wrong in the references — use the law, not the printed number.

CG / FORMULAS / COLOR & MEMORY

Color, LUT & Frame Buffer

01

Lec 2 · Graphics_SEC 3

F1

Source: Lec 2 · p23, p33–35, p39 · Graphics_SEC 3 · p2

Number of colors / gray levels

Law	Formula	Use it when
Gray levels	$levels = 2^N$	N bits per pixel (gray-scale)
Direct coding	$colors = 2^n \times 2^n \times 2^n = 2^{3n}$	n bits per primary color (R, G, B)
Bit planes	colors = 2^(number of planes)	3 planes → 8 · 6 planes → 64 · 24 planes → 16 M
Levels of each color	levels = 2^(planes per color)	e.g. 4 planes per color → 16 levels

EXAMPLE · direct coding — Graphics_SEC 3 · p2

5 bits per primary → $2^5 \times 2^5 \times 2^5 = 32 \times 32 \times 32 =$ **32,768 ≈ 32K colors**

EXAMPLE · bit planes — Graphics_SEC 3 · p2

12 planes, 4 per color → $2^4 = 16$ levels → $16 \times 16 \times 16 =$ **4,096 colors**

F2

Source: Lec 2 · p36–39

Look-up table (LUT)

Law	Formula	Use it when
LUT entries	entries = 2^(bits per pixel)	The LUT width does not change the entries
LUT size	size = (2^N × LUT width) / 8 bytes	N = pixel bits, width = bits of each entry
LUT width in bytes	size = 2^N × width (bytes)	When the width is given in bytes (e.g. "6 byte wide")

EXAMPLE · LUT — Lec 2 · p39

10-bit pixel, 24-bit LUT → entries = 2^{10} = **1024** · size = (1024 × 24)/8 = **3072 bytes**

EXAMPLE · LUT — Lec 2 · p38

8-bit pixel, 24-bit LUT → (2^8 × 24)/8 = **768 bytes**

F3

Source: Lec 2 · p38 · Graphics_SEC 3 · p2

Frame buffer memory

Law	Formula	Use it when
Frame buffer (no LUT)	size = W × H × bits per pixel / 8 bytes	Direct coding
Total with LUT	total = frame buffer + LUT size	Indexed color (LUT method)
Units	1K = 1024 · 1M = 1024 × 1024	262,144 bytes = 256K bytes

EXAMPLE · frame buffer — Graphics_SEC 3 · p2

1920 × 1080 × 32 = 66,355,200 bits / 8 = **8,294,400 bytes ≈ 8.29 MB**

EXAMPLE · system #1 vs #2 — Lec 2 · p38

1024 × 1280 × 24 bits ≈ 4 MB · 1024 × 1280 × 8 ≈ 1 MB + LUT 768 bytes

CG / FORMULAS / VIEWING

Window → Viewport & Aspect Ratio

02

Lec 3 · Graphics_Sec 4

F4

Source: Lec 3 · p5–9 · Graphics_Sec 4 · p1–2

Window to viewport mapping

Law	Formula	Use it when
Scale in x	`Sx = (XVmax − XVmin) / (XWmax − XWmin)`	Always first
Scale in y	`Sy = (YVmax − YVmin) / (YWmax − YWmin)`	Always first
World → viewport	`XV = XVmin + Sx·(XW − XWmin)` `YV = YVmin + Sy·(YW − YWmin)`	(XW, YW) given, (XV, YV) asked
Viewport → world	`XW = XWmin + (XV − XVmin)/Sx` `YW = YWmin + (YV − YVmin)/Sy`	(XV, YV) given, (XW, YW) asked
Order of the values	`(W.l, W.r, W.b, W.t) = (XWmin, XWmax, YWmin, YWmax)`	Read the window / viewport tuple

EXAMPLE · mapping — Lec 3 · p6

W (20, 80, 40, 80), V (30, 60, 40, 60), (30, 80): Sx = 30/60 = 0.5, Sy = 20/40 = 0.5 → XV = 30 + 0.5(10) = **35**, YV = 40 + 0.5(40) = **60**

EXAMPLE · inverse — Graphics_Sec 4 · p2

W (0, 10, 0, 5), V (50, 250, 100, 200), (150, 150): Sx = 20, Sy = 20 → XW = 100/20 = **5**, YW = 50/20 = **2.5**

F5

Source: Lec 3 · p8, p24–27 · Graphics_Sec 4 · p2

Aspect ratio — matching viewport

Law	Formula	Use it when
Aspect ratio	`R = width / height of the world window`	Compare R with W/H of the screen window
R > W/H	`glViewport(0, 0, W, W/R)`	Width = W, height = W/R
R < W/H	`glViewport(0, 0, H*R, H)`	Height = H, width = H·R
Default viewport	`glViewport(0, 0, W, H)`	The entire screen window

EXAMPLE · R < W/H — Lec 3 · p26

R = 1.6, 360 × 200: W/H = 1.8 > R → H = 200, W = 200 × 1.6 = 320 → **glViewport(0, 0, 320, 200)**

EXAMPLE · R > W/H — Lec 3 · p27

R = 2, 360 × 200: W/H = 1.8 < R → W = 360, H = 360/2 = 180 → **glViewport(0, 0, 360, 180)**

CG / FORMULAS / LINES

Line Drawing — DDA & Bresenham

03

Lec 4 · Graphics sec5

F6

Source: Lec 4 · p10–23 · Graphics sec5 · p1

Line equation and DDA

Law	Formula	Use it when
Slope / intercept	$m = (y_{end} - y_0)/(x_{end} - x_0)$ · $b = y_0 - m \cdot x_0$	$y = m \cdot x + b$
Differences	$\Delta x = x_{end} - x_0$ · $\Delta y = y_{end} - y_0$ · steps = $\max(\lvert\Delta x\rvert, \lvert\Delta y\rvert)$	Before any table
$\lvert m\rvert \le 1$, left → right	$x_{k+1} = x_k + 1$ · $y_{k+1} = y_k + m$	$\lvert\Delta x\rvert \ge \lvert\Delta y\rvert$, x increasing
$\lvert m\rvert \le 1$, right → left	$x_{k+1} = x_k - 1$ · $y_{k+1} = y_k - m$	x decreasing
$\lvert m\rvert > 1$, upward	$y_{k+1} = y_k + 1$ · $x_{k+1} = x_k + 1/m$	$\lvert\Delta y\rvert > \lvert\Delta x\rvert$, y increasing
$\lvert m\rvert > 1$, downward	$y_{k+1} = y_k - 1$ · $x_{k+1} = x_k - 1/m$	y decreasing
Rounding	round(a) = int(a + 0.5)	Every computed x or y
Trivial lines	$m = 0$ horizontal · $m = \infty$ vertical · $m = 1 \rightarrow y = x$	No algorithm needed

EXAMPLE · DDA — Lec 4 · p23

(9, 8) → (2, 3): $m \le 1$, steps = 7, $x - 1$, $y - m$ → (8, 7.2)→(8,7) · (7, 6.5)→(7,7) · (6, 5.8)→(6,6) …

EXAMPLE · DDA $\lvert m\rvert > 1$ — Graphics sec5 · p1

(3, 2) → (7, 8): $\lvert\Delta y\rvert > \lvert\Delta x\rvert$, $1/m = 0.667$ → (3.667, 3)→(4,3) · (4.33, 4)→(4,4) · (5, 5) …

F7

Source: Lec 4 · p28–35 · Graphics sec5 · p2

Bresenham's line algorithm

Law	Formula	Use it when
Initial decision	$p0 = 2\Delta y - \Delta x$	$0 < m < 1$ (step in x)
$p_k < 0$	next = $(x_k + 1, y_k)$ · $p_{k+1} = p_k + 2\Delta y$	y stays the same
$p_k \geq 0$	next = $(x_k + 1, y_k + 1)$ · $p_{k+1} = p_k + 2\Delta y - 2\Delta x$	y increases by 1
$\lvert\Delta y\rvert > \lvert\Delta x\rvert$	swap x and y, then use the same laws	Steep line ($m > 1$)
Repeat	$\Delta x - 1$ more times after the first point	Table: P · X · Y · SetPixel

EXAMPLE · Bresenham — Lec 4 · p30

(20, 10) → (30, 18): $\Delta x = 10$, $\Delta y = 8$, $p0 = 6$ · updates: $p - 4$ ($p \geq 0$), $p + 16$ ($p < 0$) → (21,11) (22,12) (23,12) (24,13) …

Lec 4 · p35 prints (38, 8) as the last pixel; with the laws above it is (32, 8).

CG / FORMULAS / CIRCLE & ELLIPSE

Midpoint Circle & Ellipse

04

Lec 5 · Graphics_Sec 6

F8

Source: Lec 5 · p5–14 · Graphics_Sec 6 · p1

Midpoint circle

Law	Formula	Use it when
Circle	$(x - xc)^2 + (y - yc)^2 = r^2$ · $x = xc + r \cos \theta$, $y = yc + r \sin \theta$	Direct / trigonometric form
Decision function	$P = x^2 + y^2 - R^2$	= 0 on · > 0 outside · < 0 inside
Start	$(x0, y0) = (0, r)$ · $P0 = 1 - r$	Lecture: p0 = 5/4 − r, rounded to 1 − r
$P_k < 0$	$x_{k+1} = x_k + 1$, $y_{k+1} = y_k$ · $P_{k+1} = P_k + 2x_{k+1} + 1$	Midpoint inside
$P_k \geq 0$	$x_{k+1} = x_k + 1$, $y_{k+1} = y_k - 1$ · $P_{k+1} = P_k + 2x_{k+1} + 1 - 2y_{k+1}$	Midpoint outside
Stop	$x \geq y$	One octant only
8-way symmetry	(x, y) (y, x) $(y, -x)$ $(x, -y)$ $(-x, -y)$ $(-y, -x)$ $(-y, x)$ $(-x, y)$	Other seven octants
Center ≠ (0, 0)	setPixel(xc + x, yc + y)	Move every point

EXAMPLE · circle r = 10 — Lec 5 · p12

P0 = −9 → (1,10) · −6 → (2,10) · −1 → (3,10) · 6 → (4,9) · −3 → (5,9) · 8 → (6,8) · 5 → (7,7)

F9

Source: Lec 5 · p15–23 · Graphics_Sec 6 · p2

Midpoint ellipse

Law	Formula	Use it when
Region 1 start	$(0, r_y) \cdot p1_0 = r_y^2 - r_x^2 r_y + \frac{1}{4} r_x^2$	Increment x, calculate y
$p1_k < 0$	$(x_k + 1, y_k) \cdot p1_{k+1} = p1_k + 2r_y^2 x_{k+1} + r_y^2$	Midpoint inside
$p1_k \geq 0$	$(x_k + 1, y_k - 1) \cdot p1_{k+1} = p1_k + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_y^2$	Midpoint outside
Leave region 1	$2r_y^2 x \geq 2r_x^2 y$	Check after every point
Region 2 start	last point of region 1 $\cdot p2_0 = r_y^2 (x_0 + \frac{1}{2})^2 + r_x^2 (y_0 - 1)^2 - r_x^2 r_y^2$	Decrement y, calculate x
$p2_k > 0$	$(x_k, y_k - 1) \cdot p2_{k+1} = p2_k - 2r_x^2 y_{k+1} + r_x^2$	Midpoint outside
$p2_k \leq 0$	$(x_k + 1, y_k - 1) \cdot p2_{k+1} = p2_k + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_x^2$	Midpoint inside
Stop	$y = 0$	End of region 2
Incremental	$2r_y^2 x_{k+1} = 2r_y^2 x_k + 2r_y^2 \cdot 2r_x^2 y_{k+1} = 2r_x^2 y_k - 2r_x^2$	Fill the two table columns fast
4-way symmetry	(x, y) $(-x, y)$ $(-x, -y)$ $(x, -y)$	Other three quadrants

EXAMPLE · ellipse rx = 4, ry = 3 — Graphics_Sec 6 · p2

$p1_0 = 9 - 48 + 4 = -35 \rightarrow (1,3) \cdot -8 \rightarrow (2,3) \cdot 37 \rightarrow (3,2) \cdot 36 \rightarrow (4,1)$: $72 \geq 32 \rightarrow$ region 2 · $p2_0 = 38 \rightarrow (4,0)$

EXAMPLE · ellipse rx = 8, ry = 6 — Lec 5 · p22

$p1_0 = 36 - 384 + 16 = -332$ · region 1 ends at (7, 3)

Lec 5 · p23 prints $p2_0 = -151$ for rx = 8, ry = 6; the law gives $2025 + 256 - 2304 = -23$.

CG / FORMULAS / TRANSFORMATIONS

2D / 3D Transformations

05

Lec 6 · Graphics_Sec 7

F10

Source: Lec 6 · p6–15

2D equations

Law	Formula	Use it when
Translation	x′ = x + tx · y′ = y + ty	Move by (tx, ty)
Scaling	x′ = sx·x · y′ = sy·y	About the origin
Rotation	x′ = x cos θ − y sin θ · y′ = x sin θ + y cos θ	θ > 0 counter-clockwise
Reflection	horizontal: y′ = −y · vertical: x′ = −x · diagonal: both	sy = −1 · sx = −1 · both
Shear	x-shear: x′ = x + shx·y · y-shear: y′ = shy·x + y	Shifted slices
Homogeneous	P(x, y) → P(x, y, 1) · P′ = M · P	All become multiplications

F11

Source: Lec 6 · p16–19 · Graphics_Sec 7 · p1

2D homogeneous matrices

T(tx, ty)

$$\begin{vmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{vmatrix}$$

$T^{-1} = T(-tx, -ty)$

S(sx, sy)

$$\begin{vmatrix} sx & 0 & 0 \\ 0 & sy & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$S^{-1} = S(1/sx, 1/sy)$

R(θ)

$$\begin{vmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$R^{-1}(\theta) = R(-\theta) = R^{T}(\theta)$

Horizontal reflection

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

its own inverse

Vertical reflection

$$\begin{vmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

its own inverse

Diagonal reflection

$$\begin{vmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

its own inverse

Shear in x

$$\begin{vmatrix} 1 & shx & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Shear in y

$$\begin{vmatrix} 1 & 0 & 0 \\ shy & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Rotation 90°

$$\begin{vmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$\cos 90 = 0$, $\sin 90 = 1$

Graphics_Sec 7 prints the shear inverse with 1/sh; the inverse of a shear by sh is a shear by −sh.

F12

Source: Lec 6 · p20–36 · Graphics_Sec 7 · p3

Concatenation and composite transformations

Law	Formula	Use it when
Translations add	T(tx1, ty1)·T(tx2, ty2) = T(tx1 + tx2, ty1 + ty2)	Two translations in a row
Scales multiply	S(sx1, sy1)·S(sx2, sy2) = S(sx1·sx2, sy1·sy2)	Two scales in a row
Angles add	R(θ)·R(φ) = R(θ + φ) = R(φ)·R(θ)	Two rotations in a row
Order	P′ = M_n · … · M_2 · M_1 · P	First operation is written next to P
Rotate about (a, b)	P′ = T(a, b) · R(θ) · T(−a, −b) · P	Pivot point ≠ origin
Scale about (a, b)	P′ = T(a, b) · S(sx, sy) · T(−a, −b) · P	Fixed point ≠ origin
Reflect about (a, b)	P′ = T(a, b) · S(±1, ±1) · T(−a, −b) · P	Translate, reflect, translate
Centroid	((x1 + x2 + x3)/3, (y1 + y2 + y3)/3)	Rotate about the triangle centroid

EXAMPLE · order — Lec 6 · p32

Rotate by 50°, scale double horizontally, translate 10 down → P′ = T(0, −10)·S(2, 1)·R(50)·P

EXAMPLE · pivot — Lec 6 · p33

Rotate by 40° around (3, 4) → P′ = T(3, 4)·R(40)·T(−3, −4)·P

EXAMPLE · centroid — Lec 6 · p36

A(0,0), B(2,1), C(5,4) → centroid (7/3, 5/3) = (2.33, 1.66) → T(2.33, 1.66)·R(90)·T(−2.33, −1.66)

F13

Source: Lec 6 · p37–42 · Graphics_Sec 7 · p2

3D homogeneous matrices

T(tx, ty, tz)

$$\begin{vmatrix} 1 & 0 & 0 & tx \\ 0 & 1 & 0 & ty \\ 0 & 0 & 1 & tz \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

S(sx, sy, sz)

$$\begin{vmatrix} sx & 0 & 0 & 0 \\ 0 & sy & 0 & 0 \\ 0 & 0 & sz & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

Rz(θ)

$$\begin{vmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

Rx(θ)

$$\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

Ry(θ)

$$\begin{vmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

Reflect over xy plane

$$\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

yz: −x · xz: −y